



Arming V δ 2 T Cells with Chimeric Antigen Receptors to Combat Cancer

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ABSTRACT

Immunotherapy has emerged as a promising approach in the field of cancer treatment, with chimeric antigen receptor (CAR) T-cell therapy demonstrating remarkable success. However, challenges such as tumor antigen heterogeneity, immune evasion, and the limited persistence of CAR-T cells have prompted the exploration of alternative cell types for CAR-based strategies. Gamma delta T cells, a unique subset of lymphocytes with inherent tumor recognition capabilities and versatile immune functions, have garnered increasing attention in recent years. In this review, we present how

arming V δ 2-T cells might be the basis for next-generation immunotherapies against solid tumors. Following a comprehensive overview of $\gamma\delta$ T-cell biology and innovative CAR engineering strategies, we discuss the clinical potential of V δ 2 CAR-T cells in overcoming the current limitations of immunotherapy in solid tumors. Although the applications of V δ 2 CAR-T cells in cancer research are relatively in their infancy and many challenges are yet to be identified, V δ 2 CAR-T cells represent a promising breakthrough in cancer immunotherapy.

Introduction

Immunotherapy has emerged as a promising approach in cancer treatment, with chimeric antigen receptor (CAR) T-cell therapy showing remarkable success in treating aggressive B-cell lymphomas, leading to durable remissions. However, these positive results are counterbalanced by serious life-threatening side effects of CAR-T-cell therapies, including cytokine release syndrome (CRS) and ON-target/OFF-tumor toxicity. Indeed, tumor-associated antigens (TAA) are shared with normal cells. OFF-tumor toxicity can have severe consequences, particularly when vital tissues are affected. Additionally, CAR-T cells have not succeeded similarly in treating solid tumors due to the antigenic heterogeneity, immunosuppressive tumor microenvironment, and challenges related to CAR-T-cell homing.

Although CAR engineering has primarily focused on conventional $\alpha\beta$ T cells, V δ 2-T cells hold promise as immune effectors for cancer immunotherapy. They are easy to manufacture and engineer and may be used in allogeneic settings. However, the application of V δ 2-T cells in cancer research is still in the early stages, and many challenges need to be addressed. This review provides a comprehensive overview of the V δ 2-T-cell biology and CAR engineering, to finally highlight the potential of V δ 2 CAR-T cells as the basis for next-generation immunotherapies.

V δ 2 T Cells, an Ideal Immune Effector for Cancer Immunotherapy

A bridge between innate and adaptative immunity

$\gamma\delta$ T cells, an unconventional population of T cell

To defend the organism against pathogens, the immune system includes innate and adaptative immunity. In contrast to conventional $\alpha\beta$ T cells, which are commonly discussed when referring to T cells, $\gamma\delta$ T cells break the dichotomy between innate and adaptative immunity. Indeed, they can rapidly initiate an immune response without going through the expansion and differentiation phases required for the conventional T cells to secrete a large panel of cytokines and proinflammatory chemokines and to collaborate and/or activate other cells of the immune system to increase the immune response against a pathogen [for review (1)].

In humans, $\gamma\delta$ T cells represent 3% to 5% of the T-cell population in peripheral blood and can quickly expand upon stimulation. They are the first lymphocytes to develop, even before the other immune cells become functional. Their primary role is to monitor the environment and initiate immune responses against infections, cellular transformations, and tissue damage. Their abundance varies based on factors such as age, organ, and the specific $\gamma\delta$ paired chains that form their TCR. Interestingly, their TCR is not MHC-restricted, enabling their rapid activation. They also express various activating and inhibitory NK receptors, along with different cytokine receptors like IL2R and IL15R, which finely regulate their immune activation. Once activated, $\gamma\delta$ T cells exhibit similar cytotoxic activity to $\alpha\beta$ T cells, releasing soluble mediators such as perforin and granzymes, as well as proinflammatory cytokines. They can also display a regulatory phenotype when exposed to immunosuppressive cytokines such as IL10 and TGF- β and perform immunosuppressive functions by directly secreting these cytokines. Moreover, $\gamma\delta$ T cells can function as antigen-presenting cells (APC), prime immature dendritic cells, and naive $\alpha\beta$ T cells following the phagocytosis of apoptotic or live infected or cancer cells (2). They also express classic APC markers, including MHC, CD80, CD86, and CD11a.

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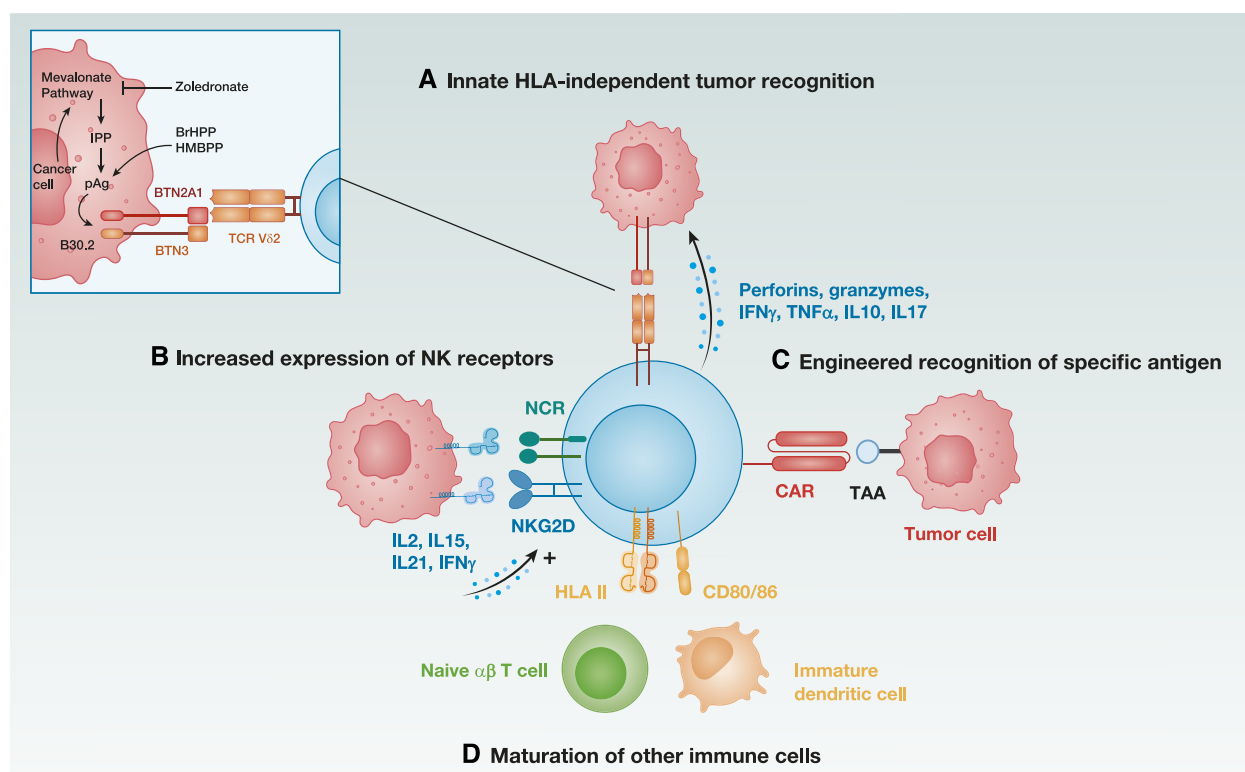


Figure 1.

Exploiting Vδ2 T cells particularities to their fullest. **A**, Vδ2 T cells present interesting innate and HLA-independent tumor recognition, sensitive to the intracellular concentration of phosphoantigen (pAg), which can be upregulated with zoledronate or IPP analogs. Several strategies can also be used to increase their efficacy, including **B**, the increase of an NK-like response with the use of different cytokines (IL2, IL15, IL21, or IFN-γ) or **C**, engineering to increase their recognition against other antigens through the use of CAR. **D**, On top of these reactivities, Vδ2 T cells exhibit CPA-like features that allow them to mature other immune cells like immature dendritic cells or naive αβ T cells. (Adapted from an image created with BioRender.com.)

Vδ2-T cells, the standard-bearer of γδ T cells

More than 90% of γδ T cells in peripheral blood express the Vδ2 chain, almost exclusively paired with the Vγ9 chain. These cells can be found at various stages of differentiation, including naive and central memory T_{CM} states when in circulation. Once at the tumor or infection site, they transition into effector T_{EM} and resident T_{RM} memory population states.

Vδ2-T cells are activated by nonpeptidic phosphorylated molecules called phosphoantigens (pAg), produced by bacteria and eukaryotic cells (Fig. 1). Among these pAgs, isopentenyl pyrophosphate (IPP) is generated from the endogenous mevalonate pathway of cholesterol synthesis. Synthetic or bacterial IPP, such as BrHPP (bromohydrin pyrophosphate) or HMBPP [(E)-1-hydroxy-2-methyl-2-butenyl-4-pyrophosphate], are more efficient at stimulating Vδ2-T cells compared with endogenous IPP. Aminobisphosphonates (ABP), which disrupt the mevalonate pathway, lead to IPP accumulation and subsequent activation of Vδ2-T cells. The discovery of the pivotal role played by butyrophilin BTN3A1 has substantially advanced our understanding of pAg-mediated activation. BTN3A1, a type-1 glycoprotein structurally similar to molecules of the B7 receptor family, is expressed on the surface of target cells. The intracellular domain of BTN3A1 is necessary for sensing pAg and forming a complex with butyrophilin BTN2A1, which results in conformational changes that enable direct interaction with the Vδ2-TCR (3–5). However, it

remains unclear whether pAg recognition follows a classic antigenic presentation or an allosteric model and the BTN3A1/BTN2A1 complex is sufficient to activate the Vδ2-TCR or it requires another unidentified ligand (3).

In addition to TCR, Vδ2-T cells express secondary receptors, which modulate their activation, including NK receptors like NKG2D and DNAM1. Upon recognizing their respective ligands, these receptors enhance the Vδ2-T-cell activation by providing a costimulatory signal. Whether these receptors can directly activate γδ T cells or if TCR engagement is required is debated. Vδ2-T cells also express inhibitory receptors such as NKG2A and immune checkpoint proteins, which play a role in immune tolerance to spare normal cells.

Vδ2-T cells, a key player in cancer immunosurveillance

Lessons from murine models

Despite the inability of murine γδ T cells to recognize pAgs, several preclinical studies have highlighted the protective role of γδ T cells in tumor development. Initial studies revealed that mice are prone to develop tumors following the genetic deletion of γδ T cells (6). Further studies demonstrated a protective effect of γδ T cells against spontaneous B-cell lymphomas (7), prostate cancer (8), and in the widely used B16-F0 melanoma model (9). In fact, a therapeutic effect was demonstrated following the adoptive transfer of human γδ T cells into mice bearing established adenocarcinomas of the prostate (8),

B16-F0 melanomas (10), and glioblastoma (11). In most studies, the protective role of $\gamma\delta$ T cells against tumor progression involved the TCR engagement and/or the NKG2D pathway (6, 11). Accordingly, NKG2D-deficient mice exhibit a higher susceptibility to cancer (12).

V δ 2-T cells, a first-line defense in cancer immune response

The innate ability of $\gamma\delta$ T cells to recognize tumor cells has garnered substantial attention as a potential approach in harnessing their potency in cancer immunotherapy. Furthermore, $\gamma\delta$ T cells have a natural tendency to infiltrate tumor sites and are frequently observed among tumor-infiltrating lymphocytes (TIL). An extensive pan-cancer study analysis using CIBERSORT revealed that $\gamma\delta$ T cells are the immune subset most substantially associated with favorable overall survival across 25 different cancer types (13). However, this study also highlighted considerable variability in tumoral $\gamma\delta$ T-cell infiltration and some challenges in distinguishing between $\gamma\delta$ T cells, NK cells, and other T-cell subsets. Further computational analysis revealed that the abundance of V δ 2-TILs is associated with a favorable outcome in various hematologic and solid tumors (14). Additionally, high circulating levels of V δ 2-T cells have been linked to reduced cancer risks and improved survival in certain patients with acute leukemia (15).

Current evaluation of V δ 2-T cells in cancer immunotherapy

Given their broad and potent recognition of tumor cells and their potential for allogeneic use, $\gamma\delta$ T cells are highly attractive immune effector candidates across a wide range of tumors and patients with diverse MHC alleles. Furthermore, abnormal activity of HMG-CoA reductase, the rate-limiting enzyme in the mevalonate pathway, has been observed in various tumors, leading to the accumulation of IPP involved in V δ 2-T-cell activation (16). Consequently, several clinical trials have sought to activate V δ 2-T cells *in vivo* by either systemically delivering ABP or *ex vivo* expanding autologous V δ 2-T cells from patient peripheral blood followed by adoptive cell transfer (ACT; Table 1; ref. 17). These strategies have been explored against multiple cancers, including multiple myeloma, non-Hodgkin lymphoma, acute myeloid leukemia (AML), prostate cancer, renal cell carcinoma, colorectal cancer, breast cancer, melanoma, and neuroblastoma (17). ABP administration is generally well-tolerated, increases the number of circulating V δ 2-T cells, and triggers various clinical responses among patients. In some cases, autologous V δ 2-T-cell ACT following *ex vivo* expansion has effectively controlled or reduced tumor progression. However, ABPs are rapidly eliminated from circulation through renal excretion and bone absorption. Additionally, the activation of V δ 2-T cells by ABP seems to induce exhaustion in preclinical primate models and clinical trials (18). Furthermore, little is known about the ability of systemic ABP to activate V δ 2-T cells within the tumor microenvironment. Finally, few clinical trials have been initiated involving an allogeneic ACT of $\gamma\delta$ T cells. Only two studies have been published, reporting the clinical safety and feasibility of this approach in patients with refractory/relapsed AML (19) and hepatocellular carcinoma (HCC) (20).

Manufacturing and engineering of V δ 2 T cells

The allogeneic potential of V δ 2 T cells allows their expansion from peripheral blood cells from healthy donors rather than patients with cancer, facilitating their manufacturing, engineering, and administration strategy, as well as the establishment of a bank which would be readily available anytime for cancer patient infusion.

Manufacturing of V δ 2 CAR-T cells

Given their broad and potent recognition of tumor cells and potential for allogeneic use, V δ 2 T cells are highly attractive immune effector candidates across a wide range of tumors and patients with diverse MHC alleles. Indeed, one of their main advantages is the establishment of a bank from healthy donors who could be readily available anytime for cancer patient infusion. V δ 2 T cells are commonly expanded from freshly isolated PBMC from healthy donors in the presence of a combination of IL2 with natural or synthetic pAg agonists, such as HMBPP or BrHPP. Globally, within 2 weeks, at least 10^8 V δ 2-T cells can be generated from whole blood-derived PBMC (<10 mL; ref. 21). These cells showed potent cytotoxic and antitumor activity *in vitro* against numerous tumor cell types. Further studies reported that the combination of pAg agonists with other cytokines, such as IL15 and IL21, enhances antitumor cytotoxicity of V δ 2 T cells (22, 23). Other protocols expanded V δ 2 T cells using zoledronate and IL2 (24). Recently, Xu and colleagues (25) developed a new V δ 2 T-cell expansion protocol involving incubation with zoledronate, IL2, IL15, and vitamin C, which resulted in improved cell proliferation, differentiation, cytotoxicity, and persistence when injected in murine xenograft models. In addition, the feeder cell-based expansion method for V δ 2 T cells has been investigated. In particular, the presence of MHC-II+ cells or the depletion of $\alpha\beta$ T cells seems to improve *in vitro* expansion of V δ 2 T cells (26, 27).

However, the success of an expansion cannot be predicted based on initial $\gamma\delta$ T-cell percentages (28), highlighting the complexity of finding optimal donors for expansion. Several studies revealed an interindividual V δ 2 subpopulation heterogeneity among healthy individuals, including variable frequency of proliferative versus cytotoxic V δ 2 functional subpopulations (29). In line with these results, we reported that *in vitro* expansion of V δ 2 T cells resulted in highly cytotoxic or poorly cytotoxic V δ 2 T cells against glioblastoma cells, depending on the donor (11). Long-term persistence of adoptively transferred V δ 2 T cells could also be a problem, as reported for the transfer of NK cells which are under the regulatory control of comparable receptors (30). Finally, although the absence of MHC restriction prevents GVHD, $\gamma\delta$ T cells express HLA molecules that can be recognized by the host immune system, which may further result in a host versus graft effect reducing $\gamma\delta$ T cells persistence and efficacy *in situ*.

CAR engineering of T cells

Most authorized CAR-T-cell therapies are developed from autologous $\alpha\beta$ T cells obtained from patients through leukapheresis. Nevertheless, *in vitro* studies have demonstrated the feasibility of CAR transduction in V δ 2-T cells and their ability to exhibit TAA-targeted cytotoxicity similar to conventional T cells (31–34). Herein, we will report the main developments in CAR-T-cell engineering.

Conventional CAR-T-cell backbone

CARs are synthetic receptors that redirect antigen recognition and trigger T-cell activation by merging an extracellular antigen-binding component with an intracellular signaling domain. Although the CAR generation is empiric, its design is key in balancing immunotherapy efficacy and toxicity (Fig. 2). CARs are commonly composed of an antigen-binding domain, which is commonly derived from the single-chain fragment variable (scFv) of an antibody, and a costimulatory and an intracellular domain, typically deriving from the CD3 ζ chain of the TCR. All domains impact CAR efficacy. For example, the scFv can induce intrinsic CAR-T-cell tonic signaling and exhaustion, whereas other studies demonstrate a close relationship

Table 1. Immunotherapy with $\gamma\delta$ T cells.

Study design	Patients	Reference	Outcomes
Clinical trials with autologous $\gamma\delta$ T cells			
Phase I, single arm, to test the safety of intra-arterial injection of $\gamma\delta$ T cells	Patients with HCC in a palliative setting	NCT00562666	Completed, no published results
Observational study of <i>ex vivo</i> expansion of $\gamma\delta$ T cells in combination with chemotherapy	Patients with ovarian cancer	NCT01606358	Completed, no published results
Exploratory study to assess the feasibility of the expansion of circulating $\gamma\delta$ T cells	Patients with newly diagnosed or relapsed/refractory AML	NCT03885076	Unknown status
Phase I, single arm, to test the safety and tolerability of the $\gamma\delta$ T cells in combination with chemotherapy	Patients with newly diagnosed glioblastoma receiving maintenance chemotherapy	NCT04165941	Not yet recruiting
Clinical trials with allogeneic $\gamma\delta$ T cells			
Phase I, single arm, to test the safety and efficacy of <i>ex vivo</i> expanded allogeneic $\gamma\delta$ T lymphocytes	Patients with active relapsed/refractory AML	NCT03790072	Completed Vydra and colleagues (1) showed the feasibility, safety, and tolerability of allogeneic haploidentical $\gamma\delta$ T cells, now progressing in a phase II clinical trial using an off-the-shelf, unrelated allogeneic donor source of $\gamma\delta$ T cells
Phase I/II, single arm, to test the efficacy of allogeneic $\gamma\delta$ T lymphocytes	Patients with advanced HCC	NCT05628545	Zhang and colleagues (2) demonstrated the clinical safety of adoptive transfer of the allogeneic $\gamma\delta$ cells in combination with locoregional ablation, with encouraging clinical efficacy against HCC
Phase I/II, single arm, to assess the safety, tolerability, and efficacy of allogeneic $\gamma\delta$ T-cell therapy	AML patients who are MRD positive	NCT05001451	Completed, no published results
Phase I, single arm, to test the maximum tolerated dose and effectiveness of donor $\gamma\delta$ T cells	Patients with AML at high risk of relapse after the allogeneic hematopoietic stem cell transplantation	NCT05886491	Active, recruiting
Phase I/II, single arm, to test the safety and impact and/or the rate of GVHD	Patients with hematologic malignancies with a partially mismatched bone marrow transplant	NCT03533816	Active, recruiting
Phase I, single arm, to test the efficacy and safety of the <i>ex vivo</i> expanded allogeneic $\gamma\delta$ T lymphocytes	Patients with refractory/relapsed AML	NCT04008381	Unknown status
Phase I, single arm, to test the safety, therapy dose, and efficacy of <i>ex vivo</i> the expanded $\gamma\delta$ T cells in combination with chemotherapy	Children with refractory, relapsed, or progressive neuroblastoma	NCT05400603	Active, recruiting
Phase I/II, single arm, to test the safety, efficacy and tolerability of the allogeneic $\gamma\delta$ T therapy	Patients with relapsed/refractory AML	NCT05886491	Not yet recruiting
Clinical trials with allogeneic CAR $\gamma\delta$ T cells			
Phase I, single arm, to test the safety, activity and safe dose of the haploidentical or allogeneic CAR-grafted $\gamma\delta$ T cells	Patients with relapsed/refractory solid tumor	NCT04107142	Completed, no published results
Phase I, single arm, to evaluate the safety, tolerability, pharmacokinetics, pharmacodynamics, and efficacy of the allogeneic CAR $\gamma\delta$ T-cell therapy	Patients with relapsed/refractory B-cell malignancies	NCT05653271	Active, recruiting
Phase I, to test the safety and efficacy of allogeneic CAR $\gamma\delta$ T therapy	Patients with relapsed/refractory B-cell malignancies	NCT04735471	Active, recruiting
Observational study to assess the long-term side effects of the allogeneic $\gamma\delta$ CAR-T-cell infusion (safety, efficacy, pharmacokinetics, and immunogenicity)	Patients with B-cell lymphoma previously enrolled in Adicet Bio study	NCT04911478	Prospective
Phase I, single arm, to evaluate the safety, tolerability and optimal dose of the allogeneic $\gamma\delta$ CAR-T-cell infusion	Patients with advanced cancers	NCT05302037	Not yet recruiting

Abbreviations: AML, acute myeloid leukemia; HCC, hepatocellular carcinoma; MRD, minimal residual disease

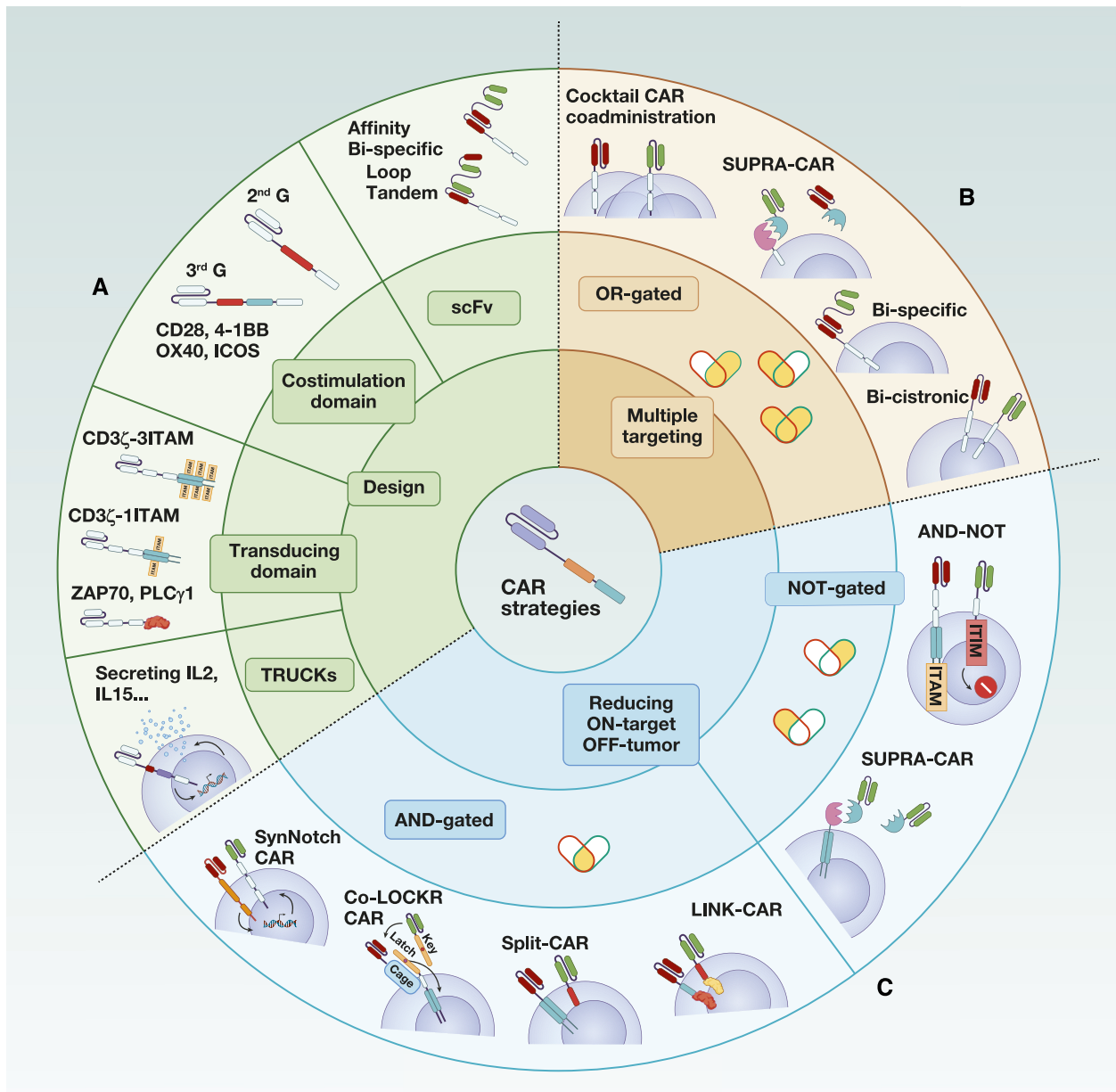


Figure 2.

Innovant strategies enhancing CAR-T-cell efficiency; CARs endow T cells with customizable antigen recognition depending on scFv specificity and antigen-binding properties. Although the generation of optimized CAR-T therapies is empiric, its design is key in balancing immunotherapy efficacy and toxicity. **A**, It relies on fine-tuning CAR affinity through scFv design, enhancing T-cell survival and persistence through the costimulatory and transducing domains. Current CAR-T-cell improvements include multiple targeting (**B**) and reducing ON-target/OFF-tumor toxicity (**C**), using various Boolean-gated strategies. Multiple targeting may be achieved by an OR-gated strategy through the coadministration of several CAR-T cell products, transduction of T cells with a bicistronic vector encoding two CARs, or use of a bispecific CAR in which several scFvs are incorporated into one CAR. To reduce the ON-target/OFF-tumor toxicity, CAR specificity may be strengthened through AND-gated CAR strategies (SynNotch-CAR, co-LOCKR-CAR, Split-CAR, or LINK-CAR) or AND-NOT-gated strategies. Finally, the SUPRA-CAR features an RR-leucine-zipper in the CAR (zipCAR) and a tumor-targeting scFv-adaptor linked to an EE-leucine-zipper (zipFv). This approach only requires the generation of various zipFvs and might be applied to various Boolean-gated strategies (OR-, NOT-, and AND-gated strategies). SUPRA-CAR: Split, universal, and programmable CAR system; TRUCKs: T-cell redirected for antigen-unrestricted cytokine-initiated killing CARs. (Adapted from an image created with BioRender.com.)

between CAR affinity, safety, and efficacy (35–38). Replacing CD3ζ with other proximal signaling molecules or altering the number of ITAM present in the CD3ζ intracellular domain improves conventional CAR-T-cell cytotoxicity and persistence (39, 40). The costimulatory domains also play a crucial role with the CD28

costimulatory domain conferring potent and immediate antitumoral cytotoxicity, whereas the 4-1BB domain enhances CAR-T-cell persistence (37, 41, 42). Whether these costimulatory domains are the best ones for Vδ2 T-cell engineering has to be investigated. Finally, CARs can also be engineered to secrete various cytokines at the

tumor site and improve their persistence and antitumor effects. These TRUCK-CARs are being evaluated in clinical trials (NCT03721068; ref. 43).

Enhancing conventional CAR-T-cell potential by Boolean logic-gating CAR

In solid tumors, CARs targeting a single antigen often prove ineffective. The main limitations of CAR immunotherapies include the heterogeneity and loss of the target antigen, as well as ON-target/OFF-tumor toxicity. Therefore, various systems have been developed for enhancing tumor targeting while minimizing normal tissue toxicity.

OR-gated strategy to widen tumor targeting: Targeting simultaneously multiple antigens can be achieved by (i) the coadministration of several CAR-T cell products, (ii) cotransduction of T cells with vectors encoding different CARs, (iii) transduction of T cells with a bicistronic vector encoding two CARs, or the use of a bispecific CAR, in which several scFvs are incorporated into one CAR (Fig. 2). Coadministration or cotransduction allows the combination of several vectors and requires minimal optimization but results in a heterogeneous mixed product with a more complex and costly manufacturing procedure. By contrast, bicistronic and bispecific CARs are cheaper to manufacture and generate more uniform products, but their design makes it more challenging to maintain their multiple functionalities (44, 45). Indeed, each scFv must not interfere with others, and the risk of cumulative side effects minimized. To encompass constant reengineering, SUPRA-CAR has been developed, offering enhanced flexibility, specificity, and controllability (46). SUPRA-CAR features an RR-leucine-zipper in the CAR (zipCAR) and a scFv-adaptor linked to an EE-leucine-zipper (zipFv). The ZipFv recognizes its specific TAA and dimerizes with the zipCAR to activate the CAR. This simplified approach of multiple targeting only requires the generation of various zipFvs.

Several preclinical studies are actively exploring multiple targeting strategies. A recent study conducted a head-to-head investigation of mono- and dual-targeting conventional CAR-T cells against BCMA and GPRC5D in a preclinical model of multiple myeloma in which they assessed the morbidity of tumor-bearing mice following the administration of either (i) BCMA-CAR and/or GPRC5D-CAR-T, (ii) bicistronic BCMA/GPRC5D CAR-T, or (iii) bispecific CAR-T cells expressing both scFvs on a single CAR (47). Although all strategies effectively eliminated tumor cells and resulted in long-term survival, the bicistronic CARs and coadministration approaches exhibited superior efficacy. Similar preclinical results have been observed in solid tumors with dual targeting of various antigen pairs, some demonstrating limited immune evasion and persistent T cells with limited exhaustion (45, 48–50). Finally, efficient targeting of three different antigens was demonstrated in several preclinical models (46, 51, 52).

The safety of multiple targeting has also been demonstrated in several clinical studies, including patients with B-cell malignancies with an initial remission rate of approximately 80% within the first 6 months. However, all patients experienced tumor relapse, despite a reduction in antigen-negative relapse (53). A phase 1 study is underway, evaluating a trispecific approach involving the coadministration of bispecific CD19/CD20 CAR-T cells with CD22-CAR-T cells.

AND-gated strategies to limit ON-target/OFF-tumor toxicity: To enhance tumor specificity, a Boolean AND-gated strategy can be achieved in which the presence of all targeted TAAs is required.

Roybal and colleagues developed a synthetic Notch receptor (syn-Notch), which, during engagement with a “gate” TAA, drives the transcription of a CAR targeting another TAA (54). The Split-CAR also restricts T-cell activity to tumor cells through combinatorial antigens recognition because the activation and costimulatory domains are provided by two distinct structures, each targeting a different TAA (55–57). Split-CARs have been further refined into co-LOCKR-CARs by exploiting the reversible interaction between two colocalized proteins (58). Lastly, exploiting the fine dissection of intracellular proximal TCR signaling molecules, a LINK-CAR has been developed, which incorporates LAT and SLP76 fused to CD28 and CD8. Although all CAR-T cells efficiently eradicate tumor cells without toxicity to normal tissues, the LINK-CAR outperforms all other Boolean-gated CARs in ROR1⁺-tumor models, one of the best preclinical models to investigate ON-target/OFF-tumor toxicity (40, 59).

An alternative approach to enhance tumor specificity is to use an AND–NOT-gated strategy, often referred to as NOT-gated, in which a CAR is expressed with an inhibitory CAR targeting an antigen present on healthy tissues but not on the tumor (Fig. 2). Although this strategy is barely investigated, a proof-of-concept has been recently provided (60).

Current Vδ2 CAR-T-cell approaches

CAR engineering in Vδ2-T cells was reported 2 decades ago (61). Vδ2-T cells were expanded from the peripheral blood of healthy donors and efficiently transduced with recombinant retrovirus encoding GD2- or CD19-CARs. These Vδ2 CAR-T cells efficiently recognized and killed antigen-expressing tumor cells (61). Unlike CD19-CAR conventional T cells, CD19-CAR Vδ2 T cells were also able to kill CD19-negative leukemia cells when the endogenous production of pAg was enhanced by ABP or tumor cells lacked MHC-I expression (32, 62). Recently, Vδ2 CAR-T cells generated from iPSC demonstrated sustained tumor cytotoxicity in the presence of IL15 (63). Finally, we recently demonstrated that Vδ2 CAR-T cells exhibited similar cytotoxicity to conventional CAR-T cells against pediatric high-grade glioma (BioRxiv 2023.11.17. 567375v1). Notably, we also highlighted their promising allogeneic settings using 3D tumor models.

Two clinical approaches of ACT with allogeneic Vδ2 CAR-T cells have recently been initiated (Table 1). Most studies include patients with B-cell malignancies and investigate ACT of allogeneic Vδ2-T cells expressing anti-CD20-CAR. Initial results reported that these Vδ2 CAR-T cells were well-tolerated, with a favorable safety profile and encouraging preliminary efficacy (64). The other approach, conducted in Singapore, investigates the safety and tolerability of NKG2L-CAR Vδ2-T cells in patients with relapsed or refractory tumors of different types. No published results are currently available.

Collectively, these data have created considerable enthusiasm about the clinical application of Vδ2 CAR-T cells, particularly in an allogeneic setting. In particular, the application of conventional CAR engineering to these cells should improve immunotherapy efficacy. Although further development is required for Boolean logic-gating CAR with conventional T cells, such strategies should also be applied to Vδ2-T-cell immunotherapy. Recently, Vδ1-T cells have also been explored for therapeutic purposes owing to the recent development of a safe clinical-grade protocol to expand them (65). These DOT cells (Delta One T cells) showed potent and specific cytotoxicity against AML and Glioblastoma cells *in vitro*. They recently entered clinical testing in patients with AML (NCT05001451; Table 1).

Vδ2 CAR-T cells as the basis for next-generation immunotherapies against solid tumors

Most current immunotherapies utilize the patient's immune system to eradicate cancer and involve *ex vivo* expansion of patient lymphocytes (TIL) before ACT. Given the ability to engineer Vδ2 T cells using Vδ2 CAR-T rather than the conventional αβ CAR-T cells offers numerous advantages, in particular, (i) an innate OR-gated strategy through their TCR, NKG2D diversity, and/or CAR engineering and (ii) an allogeneic setting of ACT and off-the-shelf CAR-T cells (Table 2).

Multiple immunoreactivities of Vδ2 CAR-T cells to overcome tumor heterogeneity

Solid tumors are characterized by strong inter- and intra-patient heterogeneities that definitively limit the actual CAR-T-cell therapies, which is even reinforced by the loss of the targeted TAA expression following CAR-T-cell therapy. In this context, using Vδ2 CAR-T rather than the conventional αβ CAR-T cells may prevent tumor escape caused by antigen loss or downregulation observed with conventional CAR-T-cell therapy. Indeed, Vδ2 CAR-T cells can recognize a myriad of TAAs through their TCR, and NKG2D receptors. This diversity of ligands combined with their differential regulation makes Vδ2 T cells sensitive to a wider range of tumor signals and more alert than αβ T cells to tumor immune evasion (66). As mentioned above, tumor cells exhibit a deregulated mevalonate pathway leading to Vδ2 T-cell activation (16). Numerous studies have reported that BTN3A1, along with various NKG2D ligands, are commonly expressed in cancer (11, 34, 67, 68). The importance of the NKG2D pathway in tumor immunosurveillance has been demonstrated through preclinical and clinical studies involving conventional NKG2D-CAR-T cells (50, 69, 70). These cells can migrate to tumor sites, extending the survival of tumor-bearing mice and providing protection against tumor rechallenge (71). Notably, some tumor cells shed NKG2D ligands from their surface to evade immunosurveillance (67, 72). Accordingly, elevated serum concentrations of shed MICA have been associated with disease progression in numerous human cancers (73). Notably, inhibition of this shedding with specific antibodies has shown a substantial survival benefit in humanized metastasis models (73).

This innate immunoreactivity against a wide range of tumor cells from patients with diverse MHC alleles may be further complemented with CAR engineering, as with conventional CAR-T cells. As stated above, preclinical studies have reported a dual reactivity of Vδ2 CD19-CAR-T cells, engaging their CAR and TCR against leukemia cells in the presence of ABP (32). This dual reactivity was confirmed with the demonstration that CAR and TCR form distinct signaling synapses, and even collaborate in CAR-T-cell activation (74). Interestingly, Vδ2 CD19-CAR-T cells were also able to kill CD19-negative leukemia cells when the endogenous production of pAg was enhanced by ABP (32). Consequently, multiple groups are currently working to improve the biodistribution of ABP to enhance Vδ2-CAR-T-cell efficacy. The current urgent need is the identification of relevant TAAs. (for review see ref. 75). TRUCK-CAR engineering might also be important to consider in improving the Vδ2 T-cell amplification and/or immunoreactivity.

Limiting the immunosuppressive tumor microenvironment with Vδ2 CAR-T cells

Vδ2 T cells are actively recruited within the tumor microenvironment (TME) and have the potential to kill tumor cells. However,

Table 2. Comparison of Vδ2 CAR-T-cell strategies with conventional CAR-T cells.

Immune features	αβ T	Vδ2 T
Innate immunity		
Immunosurveillance	No	Yes
Early cytotoxicity	No	Yes
APC	No	Yes
Adaptative immunity		
Delayed cytotoxicity	Yes	No
Memory phenotype	Yes	Yes
Antigenic diversity	High	Low
Tumor-targeting properties		
Innate recognition	No	Yes
Exhaustion	Medium	High
Regulatory phenotype	Yes	Yes
Tumor homing	Low/medium	Low/medium
ACT manufacturing and settings		
Autologous	Yes	Yes
Allogeneic	No	Yes
Amplification	Yes	Yes
CAR engineering	Yes	Yes

TME also attenuates Vδ2 T-cell responses in multiple ways. Thus, a major challenge is to limit this TME-induced immunosuppression.

Cytokinic immunosuppressive microenvironment

The TME of solid tumors is mainly dominated by proinflammatory cytokines, such as TNF-α, IL1, and IL6, which promotes tumor cell proliferation and recruit immunosuppressive cells including regulatory T cells (Treg), M2-polarized tumor-associated macrophages (M2-TAM), and immature myeloid cells (MDSC). These cells produced various immunosuppressive cytokines, including TGF-β, IL4, and IL10 involved in tumor escape. Accumulation of Treg and M2-TAM also correlates with a poor prognosis. Accordingly, the depletion of endogenous lymphocytes through chemotherapy improves ACT clinical responses (76, 77). Thus, chemotherapy or radiotherapy might be required for optimal clinical responses of ACT in patients who can support such conditioning regimens.

TRUCK-CAR engineering should also be considered to improve Vδ2-T-cell amplification and/or immunoreactivity. Indeed, several studies revealed that the infusion of conventional CAR-T cells secreting IL12 or IL18 prevents Treg- and M2-TAM immunosuppression while promoting their activation and recruitment (77–79). CAR-T cells secreting IL7 and CCL19 also showed superior tumor suppression to conventional CAR-T cells in several murine models (80, 81). Based on these results, a phase 1 clinical trial has been initiated on patients with advanced carcinoma (NCT03198546). All these strategies currently developed using conventional T cells can be applied to Vδ2 T cells. Finally, as potent activators of Vδ2-T cells, IL15 and IL21 could be considered to be secreted by Vδ2 CAR-T cells (22, 23). A major concern with IL15 or IL21 is an enhanced proliferation of Vδ2 CAR-T cells leading to the production of toxic levels of IFNγ and TNFα, which might favor the incidence of CRS, a potential fatal adverse effect of CAR-T-cell immunotherapy. In this context, the generation of Vδ2 CAR-T cells secreting CK in an inducible manner may guarantee the safety of this strategy.

Metabolic immunosuppressive microenvironment

TME exhibits abnormal metabolic characteristics, including hypoxia, acidic conditions, and limited glucose availability, because of

Table 3. Clinical trials with autologous CAR-T cells expressing ICIs in solid tumors.

Study design	Sponsors (Reference)
CTLA4 and PD1 antibodies expressing EGFR-CAR-T cells for EGFR positive advanced solid tumor	Shanghai Cell Therapy Research Institute, China (NCT03182816, NCT03182803), Shanghai International Medical Center, China (NCT02862028)
PD1 antibody expressing CAR-T cells for mesothelin positive advanced malignancies	Ningbo Cancer Hospital, China (NCT03030001, NCT02873390)
Anti-MUC1 CAR-T cells and PD1 knockout engineered T cells for NSCLC	The First Affiliated Hospital of Guangdong Pharmaceutical University, China (NCT03525782)
CAR-T and PD1 knockout engineered T cells for esophageal cancer	The First Affiliated Hospital of Guangdong Pharmaceutical University, China (NCT03706326)
CTLA4 and PD1 antibodies expressing MUC1-CAR-T cells for MUC1 positive advanced solid tumor	Ningbo Cancer Hospital, China (NCT03179007)
PD1 knockout anti-MUC1 CAR-T cells in the treatment of advanced breast cancer	Sun Yat-Sen Memorial Hospital of Sun Yat-Sen University, China (NCT05812326)
PD1 antibody expressing mesoCAR-T cells for mesothelin positive advanced solid tumor	Shanghai Cell Therapy Research Institute, China (NCT03615313)
αPD1-MSLN-CAR-T cells for the treatment of MSLN-positive advanced solid tumors	Wuhan Union Hospital, China (NCT04489862), Shanghai Mengchao Cancer Hospital, China (NCT05373147), Shanghai Cell Therapy Group Co., Ltd., China (NCT04503980, NCT05089266)
Study of CRISPR-Cas9 mediated PD1 and TCR gene-knocked out mesothelin-directed CAR-T cells in patients with mesothelin positive multiple solid tumors	Chinese PLA General Hospital, China (NCT03545815)
Study of PD1 gene-knocked out mesothelin-directed CAR-T cells with the conditioning of PC in mesothelin positive multiple solid tumors	Chinese PLA General Hospital, China (NCT03747965)
PD1 silent PSMA/PSCA targeted CAR-T for the treatment of prostate cancer	Shanghai Unicar-Therapy Bio-medicine Technology Co., Ltd., China (NCT05732948)
Clinical study on the efficacy and safety of c-Met/PD-L1 CAR-T-cell injection in the treatment of HCC	The Second Hospital of Nanjing Medical University, China (NCT03672305)
Dual-targeting CLDN18.2 and PD-L1 CAR-T for patients with CLDN18.2-positive advanced solid tumors	Sichuan University, China (NCT06084286)
Anti-PD-L1 armored anti-CD22 CAR-T/CAR-TILs targeting patients with solid tumors	Hebei Senlang Biotechnology Inc., Ltd., China (NCT04556669)
Dual-targeting HER2 and PD-L1 CAR-T for cancers with pleural or peritoneal metastasis	Sichuan University, China (NCT04684459)
Dual-targeting VEGFR1 and PD-L1 CAR-T for cancer patients with pleural or peritoneal metastases	Sun Yat-Sen Memorial Hospital of Sun Yat-Sen University, China (NCT05812326)

dysfunctional blood vessels and the high nutrient demand of tumor cells. This metabolic environment hampers T-cell function while promoting Treg differentiation and M2-TAM polarization (82–85). Hypoxia further reduces Vδ2-T-cell cytotoxicity and persistence while enhancing Treg activity (86–88). L-arginine and lipid metabolism also play vital roles in modulating Vδ2-T-cell activity (89).

In this context, metabolic inhibitors may improve Vδ2-T-cell function. The hyperactive mevalonate metabolism in tumor cells may counterbalance the reduction in Vδ2-T-cell cytotoxicity due to LDL binding on their receptors upon activation (90). Metformin, which preferentially inhibits mitochondrial oxidative phosphorylation in tumor cells, effectively reduces tumor hypoxia and specifically restores Vδ2-T-cell cytotoxicity against tumor cells in preclinical models (87). Additionally, PI3K inhibitors reduce Treg and M2-TAM polarization while improving CAR-T-cell efficacy and persistence in preclinical studies leading to its clinical evaluation (91, 92).

To optimize the Vδ2 CAR-T-cell functionality, CAR design is crucial for achieving rapid tumor regression and sustained immunosurveillance. A dual CAR design incorporating CD28 and 4-1BB costimulatory domains holds promise, especially in cases of heterogeneous tumor antigen expression (93). Finally, the genetic engineering of CAR-T cells can be harnessed to confer metabolic advantages, as demonstrated with enzymes involved in arginine metabolism (94).

Recruitment of Vδ2 CAR-T cells

Patient data and *in vivo* imaging revealed that CAR-T cells primarily circulate in the bloodstream but struggle to infiltrate solid tumors because of their dense ECM and cancer-associated fibroblasts (CAF; ref. 95). Through the production of the ECM, CAFs are a shell that traps T cells and inhibits their homing to and infiltration within the tumor (95). Depletion of CAF enhances antitumor immunity in murine models but was not successful in clinics (96). Reexpression of heparanase in engineered T cells enhances CAR-T-cell infiltration, improving survival in murine models (97). Heparanase may also improve tumor homing and infiltration by facilitating chemokine release. Excluded cytotoxic T cells also upregulate CTLA4, which modifies integrin expression reducing their migration. Accordingly, anti-CTLA4 antibodies boost T-cell motility (98) but also increase Vδ2-T-cell frequency in some patients, adding another benefit of Vδ2 CAR-T-cell immunotherapy (99).

Chemokines and cytokines, such as TGF-β and soluble NKG2D ligands secreted by CAFs, are also interfering with the Vδ2 T-cell proliferation, function, and recruitment (100). Mismatched chemokines and TCR also hinder CAR-T-cell homing. Tumor-produced CCL22 or CCL28 favor Treg recruitment and M2-TAM polarization, leading to TGF-β secretion, CAF differentiation, and T-cell exclusion. Interestingly, blocking NOX4, a TGF-β

downstream target, reverses CAF differentiation and promotes T-cell infiltration (95) while reducing TGF- β interference with V δ 2-T-cell function (100). Expression of specific receptors (CXCR2, CCR2, and CCR4) and chemokines (IL7 and CCL19) by CAR-T cells also enhances tumor homing and infiltration without compromising cytotoxicity (80, 101). It is also noteworthy that IPP released by tumor cells treated with ABP can act as a chemotactic molecule on V δ 2 T cells, a function that could help their tumor homing, even though this has yet to be confirmed *in vivo*.

Considering the challenging weak immune cell infiltration into tumors, the mode of CAR-T-cell administration is crucial. Clinical trials have shown that locoregional delivery is a safe and effective approach (NCT00730613 and NCT01082926). Additionally, engineered biomaterial delivery systems for CAR-T cells are under investigation. For example, bioactive polymer implants acting as reservoirs, releasing CAR-T cells as the material biodegrades or bioactive carriers delivering, expanding, and dispersing tumor-reactive T cells have been developed (102, 103). Both approaches have shown better tumor regression in preclinical models compared with systemic lymphocyte injections. These implants can also release vaccine adjuvants to activate robust T-cell responses. For example, combining CAR-T cells with STING agonists in biomaterial implants triggers a synergistic activation of the host APC and CAR-T cells, leading to local tumor elimination, systemic antitumor immunity, and protection against tumor rechallenge (103). This approach could potentially be employed to deliver additional therapies, such as ABP, to reinvigorate V δ 2 CAR-T cells, enhance their infiltration, and boost their immunoreactivity.

Combining V δ 2 CAR-T cells with conventional treatments

Combination with standard-of-care therapies

Radiotherapy remains a standard cancer treatment, including against solid tumors. This treatment can trigger various potent antitumor immune responses, including unmasking hidden tumor antigens, generating neoantigens, and inducing the expression of NKG2D ligands (104). A recent study highlighted the synergistic effect of combining radiotherapy with conventional NKG2D-CAR-T cells in a preclinical model (71). This combination enhanced CAR-T-cell homing to the tumor site and increased their cytotoxicity, leading to long-term protection against tumor rechallenge. Retrospective analyses have also indicated that radiotherapy can improve local tumor control (105). Similar benefits have been observed with chemotherapy, which not only induces the rapid infiltration of $\gamma\delta$ T cells into the tumor but also results in a substantial increase in tumor cell death in various preclinical models (106–108). Engineered drug-resistant V δ 2-T cells have even been designed to prevent their elimination via chemotherapy (106). The feasibility of combining this approach with V δ 2 CAR-T cells is currently under investigation in a clinical setting, and the initial results are yet to be reported (NCT04165941).

Combination with immune checkpoint inhibitors

The discovery of immune checkpoint proteins and their role in tumor tolerance and T-cell exhaustion has led to substantial advancements in cancer immunotherapy. Clinical trials have been exploring the use of immune checkpoint inhibitors (ICI) to reinvigorate

preexisting antitumor T-cell responses (109–111). Treatments with ICIs, such as PD1, PDL1, and CTLA4 inhibitors, have been approved for various cancer types and have shown promise in reactivating the immune response against tumors (111). Combining ICI with CAR-T-cell therapies is under active clinical evaluation with promising results (NCT02706405, NCT03310619, NCT02926833, and NCT03287817). However, many patients do not respond to ICI treatment and experience tumor relapse within 2 years. In this context, V δ 2 CAR-T cells could be designed to secrete antibodies against immune checkpoints like PDL1 or PD1, a strategy that has shown promise in preclinical studies with conventional CAR-T cells. A recent clinical study, involving CD19/PD1-CAR-T cells in 17 patients with PD-L1 + B lymphoma, has demonstrated the feasibility and potential efficacy of this approach (111). As a result, numerous clinical trials are currently recruiting to explore the potential of autologous CAR-T cells, which express immune checkpoint antibodies (Table 3). Interestingly, in a model of EBV-driven B-cell lymphoma, whereas conventional T-cell responses were held in check by inhibitory ligands and required coadministration of anti-PD1 and anti-CTLA4 antibodies to reduce tumor burden, a single dose of V δ 2 T cells almost completely prevented the outgrowth of tumors (112).

Conclusions

Recent advances in cancer immunotherapy have revolutionized the treatment of several cancers. However, further strategies are definitively required to fully realize the potential of cancer immunotherapy. V δ 2-T cells, with their unique characteristics, present promising opportunities for cancer immunotherapy. The combination of the antitumor-directed V δ 2 TCR with the combination of intrinsic NK cell properties and CAR engineering make them attractive candidates for competing solid tumors. These features are reinforced by their potential for allogeneic use and banking feasibility. Previously, clinical trials have shown that the ACT of autologous V δ 2-T cells is well-tolerated and has the potential to trigger antitumor immune responses. Ongoing clinical trials will provide crucial results to determine the benefits of CAR-engineered V δ 2-T cells. These results will be instrumental in shaping the future of cancer immunotherapy and may open new avenues for the treatment of solid tumors.

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References

1. Ribot JC, Lopes N, Silva-Santos B. $\gamma\delta$ T cells in tissue physiology and surveillance. *Nat Rev Immunol* 2021;21:221–32.
2. Himoudi N, Morgenstern DA, Yan M, Vernay B, Saraiva L, Wu Y, et al. Human $\gamma\delta$ T lymphocytes are licensed for professional antigen

- presentation by interaction with opsonized target cells. *J Immunol* 2012; 188:1708–16.
3. Karunakaran MM, Willcox CR, Salim M, Paletta D, Fichtner AS, Noll A, et al. Butyrophilin-2A1 directly binds germline-encoded regions of the V γ 9V δ 2 TCR and is essential for phosphoantigen sensing. *Immunity* 2020;52:487–98. e6.
 4. Rigau M, Ostrouska S, Fulford TS, Johnson DN, Woods K, Ruan Z, et al. Butyrophilin 2A1 is essential for phosphoantigen reactivity by $\gamma\delta$ T cells. *Science* 2020;367:eaay5516.
 5. Sandstrom A, Peigné C-M, Léger A, Crooks JE, Konczak F, Gesnel M-C, et al. The intracellular B30.2 domain of butyrophilin 3A1 binds phosphoantigens to mediate activation of human V γ 9V δ 2 T cells. *Immunity* 2014;40:490–500.
 6. Girardi M, Oppenheim DE, Steele CR, Lewis JM, Glusac E, Filler R, et al. Regulation of cutaneous malignancy by gammadelta T cells. *Science* 2001; 294:605–9.
 7. Street SEA, Hayakawa Y, Zhan Y, Lew AM, MacGregor D, Jamieson AM, et al. Innate immune surveillance of spontaneous B cell lymphomas by natural killer cells and gammadelta T cells. *J Exp Med* 2004;199:879–84.
 8. Liu Z, Eltoum I-EA, Guo B, Beck BH, Cloud GA, Lopez RD. Protective immunosurveillance and therapeutic antitumor activity of gammadelta T cells demonstrated in a mouse model of prostate cancer. *J Immunol* 2008;180: 6044–53.
 9. Gao Y, Yang W, Pan M, Scully E, Girardi M, Augenlicht LH, et al. Gamma delta T cells provide an early source of interferon gamma in tumor immunity. *J Exp Med* 2003;198:433–42.
 10. Lança T, Costa MF, Gonçalves-Sousa N, Rei M, Grosso AR, Penido C, et al. Protective role of the inflammatory CCR2/CCL2 chemokine pathway through recruitment of type 1 cytotoxic $\gamma\delta$ T lymphocytes to tumor beds. *J Immunol* 2013;190:6673–80.
 11. Chauvin C, Joalland N, Perroteau J, Jarry U, Lafrance L, Willem C, et al. NKG2D controls natural reactivity of V γ 9V δ 2 T lymphocytes against mesenchymal glioblastoma cells. *Clin Cancer Res* 2019;25:7218–28.
 12. Guerra N, Tan YX, Joncker NT, Choy A, Gallardo F, Xiong N, et al. NKG2D-deficient mice are defective in tumor surveillance in models of spontaneous malignancy. *Immunity* 2008;28:571–80.
 13. Tosolini M, Pont F, Poupot M, Vergez F, Nicolau-Travers M-L, Vermijlen D, et al. Assessment of tumor-infiltrating TCRV γ 9V δ 2 $\gamma\delta$ lymphocyte abundance by deconvolution of human cancers microarrays. *Oncoimmunology* 2017;6:e1284723.
 14. Gentles AJ, Newman AM, Liu CL, Bratman SV, Feng W, Kim D, et al. The prognostic landscape of genes and infiltrating immune cells across human cancers. *Nat Med* 2015;21:938–45.
 15. Godder KT, Henslee-Downey PJ, Mehta J, Park BS, Chiang K-Y, Abhyankar S, et al. Long term disease-free survival in acute leukemia patients recovering with increased gammadelta T cells after partially mismatched related donor bone marrow transplantation. *Bone Marrow Transpl* 2007;39:751–7.
 16. Gober H-J, Kistowska M, Angman L, Jenö P, Mori L, De Libero G. Human T cell receptor gammadelta cells recognize endogenous mevalonate metabolites in tumor cells. *J Exp Med* 2003;197:163–8.
 17. Hoeres T, Smetak M, Pretscher D, Wilhelm M. Improving the efficiency of V γ 9V δ 2 T-cell immunotherapy in cancer. *Front Immunol* 2018;9:800.
 18. Lin JH. Bisphosphonates: a review of their pharmacokinetic properties. *Bone* 1996;18:75–85.
 19. Vydra J, Cosimo E, Lesný P, Wanless RS, Anderson J, Clark AG, et al. A phase I trial of allogeneic $\gamma\delta$ T lymphocytes from haploidentical donors in patients with refractory or relapsed acute myeloid leukemia. *Clin Lymphoma Myeloma Leuk* 2023;23:e232–9.
 20. Zhang T, Chen J, Niu L, Liu Y, Ye G, Jiang M, et al. Clinical safety and efficacy of locoregional therapy combined with adoptive transfer of allogeneic $\gamma\delta$ T cells for advanced hepatocellular carcinoma and intrahepatic cholangiocarcinoma. *J Vasc Interv Radiol* 2022;33:19–27.e3.
 21. Kondo M, Sakuta K, Noguchi A, Ariyoshi N, Sato K, Sato S, et al. Zoledronate facilitates large-scale ex vivo expansion of functional gammadelta T cells from cancer patients for use in adoptive immunotherapy. *Cytotherapy* 2008;10:842–56.
 22. Thedrez A, Harly C, Morice A, Salot S, Bonneville M, Scotet E. IL-21-mediated potentiation of antitumor cytolytic and proinflammatory responses of human V gamma 9V delta 2 T cells for adoptive immunotherapy. *J Immunol* 2009;182:3423–31.
 23. Aehnlich P, Carnaz Simões AM, Skadborg SK, Holmen Olofsson G, Thor Straten P. Expansion with IL-15 increases cytotoxicity of V γ 9V δ 2 T cells and is associated with higher levels of cytotoxic molecules and T-bet. *Front Immunol* 2020;11:1868.
 24. Kondo M, Izumi T, Fujieda N, Kondo A, Morishita T, Matsushita H, et al. Expansion of human peripheral blood $\gamma\delta$ T cells using zoledronate. *J Vis Exp* 2011;55:3182.
 25. Xu Y, Xiang Z, Alnaggar M, Kouakanou L, Li J, He J, et al. Allogeneic V γ 9V δ 2 T-cell immunotherapy exhibits promising clinical safety and prolongs the survival of patients with late-stage lung or liver cancer. *Cell Mol Immunol* 2021;18:427–39.
 26. Soriano-Sarabia N, Sandvold H, Jomaa H, Kubin T, Bein G, Hackstein H. Primary MHC-class II(+) cells are necessary to promote resting V δ 2 cell expansion in response to (E)-4-hydroxy-3-methyl-but-2-enyl-pyrophosphate and isopentenyl pyrophosphate. *J Immunol* 2012;189:5212–22.
 27. Landin AM, Cox C, Yu B, Bejanyan N, Davila M, Kelley L. Expansion and enrichment of gamma-delta ($\gamma\delta$) T cells from apheresed human product. *J Vis Exp* 2021;175.
 28. Burnham RE, Zoine JT, Story JY, Garimalla SN, Gibson G, Rae A, et al. Characterization of donor variability for $\gamma\delta$ T cell *ex vivo* expansion and development of an allogeneic $\gamma\delta$ T cell immunotherapy. *Front Med Lausanne* 2020;7:588453.
 29. Ryan PL, Sumaria N, Holland CJ, Bradford CM, Izotova N, Grandjean CL, et al. Heterogeneous yet stable V δ 2(+) T-cell profiles define distinct cytotoxic effector potentials in healthy human individuals. *Proc Natl Acad Sci U S A* 2016;113:14378–83.
 30. de Witte MA, Sarhan D, Davis Z, Felices M, Vallera DA, Hinderlie P, et al. Early reconstitution of NK and $\gamma\delta$ T cells and its implication for the design of post-transplant immunotherapy. *Biol Blood Marrow Transplant* 2018;24: 1152–62.
 31. Fleischer LC, Becker SA, Ryan RE, Fedanov A, Doering CB, Spencer HT. Non-signaling chimeric antigen receptors enhance antigen-directed killing by $\gamma\delta$ T cells in contrast to $\alpha\beta$ T cells. *Mol Ther Oncolytics* 2020;18:149–60.
 32. Rozenbaum M, Meir A, Aharon Y, Itzhaki O, Schachter J, Bank I, et al. Gamma-delta CAR-T cells show CAR-directed and independent activity against leukemia. *Front Immunol* 2020;11:1347.
 33. Zhai X, You F, Xiang S, Jiang L, Chen D, Li Y, et al. MUC1-Tn-targeting chimeric antigen receptor-modified V γ 9V δ 2 T cells with enhanced antigen-specific anti-tumor activity. *Am J Cancer Res* 2021;11:79–91.
 34. Capsomidis A, Benthall G, Van Acker HH, Fisher J, Kramer AM, Abeln Z, et al. Chimeric antigen receptor-engineered human gamma delta T cells: enhanced cytotoxicity with retention of cross presentation. *Mol Ther* 2018;26: 354–65.
 35. Richman SA, Nunez-Cruz S, Moghimi B, Li LZ, Gershenson ZT, Mourelatos Z, et al. High-affinity GD2-specific CAR T cells induce fatal encephalitis in a preclinical neuroblastoma model. *Cancer Immunol Res* 2018;6:36–46.
 36. Mount CW, Majzner RG, Sundaresh S, Arnold EP, Kadapakkam M, Haile S, et al. Potent antitumor efficacy of anti-GD2 CAR T-cells in H3K27M⁺ diffuse midline gliomas. *Nat Med* 2018;24:572–9.
 37. Long AH, Haso WM, Shern JF, Wanhainen KM, Murgai M, Ingaramo M, et al. 4-1BB costimulation ameliorates T cell exhaustion induced by tonic signaling of chimeric antigen receptors. *Nat Med* 2015;21:581–90.
 38. Majzner RG, Theruvath JL, Nellan A, Heitzeneder S, Cui Y, Mount CW, et al. CAR T cells targeting B7-H3, a pan-cancer antigen, demonstrate potent preclinical activity against pediatric solid tumors and brain tumors. *Clin Cancer Res* 2019;25:2560–74.
 39. Feucht J, Sun J, Eyquem J, Ho Y-J, Zhao Z, Leibold J, et al. Calibration of CAR activation potential directs alternative T cell fates and therapeutic potency. *Nat Med* 2019;25:82–8.
 40. Tousley AM, Rotiroti MC, Labanieh L, Rysavy LW, Kim W-J, Lareau C, et al. Co-opting signalling molecules enables logic-gated control of CAR T cells. *Nature* 2023;615:507–16.
 41. Zhao Z, Condomines M, van der Stegen SJC, Perna F, Kloss CC, Gunset G, et al. Structural design of engineered costimulation determines tumor rejection kinetics and persistence of CAR T cells. *Cancer Cell* 2015;28:415–28.
 42. Kawalekar OU, O'Connor RS, Fraietta JA, Guo L, McGettigan SE, Posey AD, et al. Distinct signaling of coreceptors regulates specific metabolism pathways and impacts memory development in CAR T cells. *Immunity* 2016;44: 380–90.
 43. Chen Y, Sun C, Landoni E, Metelitsa L, Dotti G, Savoldo B. Eradication of neuroblastoma by T cells redirected with an optimized GD2-specific chimeric antigen receptor and interleukin-15. *Clin Cancer Res* 2019;25:2915–24.

44. Tong C, Zhang Y, Liu Y, Ji X, Zhang W, Guo Y, et al. Optimized tandem CD19/CD20 CAR-engineered T cells in refractory/relapsed B-cell lymphoma. *Blood* 2020;136:1632–44.
45. Tian M, Cheuk AT, Wei JS, Abdelmaksoud A, Chou H-C, Milewski D, et al. An optimized bicistronic chimeric antigen receptor against GPC2 or CD276 overcomes heterogeneous expression in neuroblastoma. *J Clin Invest* 2022; 132:e155621.
46. Cho JH, Collins JJ, Wong WW. Universal chimeric antigen receptors for multiplexed and logical control of T cell responses. *Cell* 2018;173:1426–38. e11.
47. de Larrea CF, Staehr M, Lopez AV, Ng KY, Chen Y, Godfrey WD, et al. Defining an optimal dual-targeted CAR T-cell therapy approach simultaneously targeting BCMA and GPRC5D to prevent BCMA escape-driven relapse in multiple myeloma. *Blood Cancer Discov* 2020;1:146–54.
48. Schmidts A, Srivastava AA, Ramapriyan R, Bailey SR, Bouffard AA, Cahill DP, et al. Tandem chimeric antigen receptor (CAR) T cells targeting EGFRvIII and IL-13Ra2 are effective against heterogeneous glioblastoma. *Neurooncol Adv* 2022;5:vdac185.
49. Chen C, Li K, Jiang H, Song F, Gao H, Pan X, et al. Development of T cells carrying two complementary chimeric antigen receptors against glypican-3 and asialoglycoprotein receptor 1 for the treatment of hepatocellular carcinoma. *Cancer Immunol Immunother* 2017;66:475–89.
50. Li S, Zhao R, Zheng D, Qin L, Cui Y, Li Y, et al. DAP10 integration in CAR-T cells enhances the killing of heterogeneous tumors by harnessing endogenous NKG2D. *Mol Ther Oncolytics* 2022;26:15–26.
51. Vasu S, Alinari L, Szuminski N, Schneider D, Denlinger N, Chan WK, et al. A phase I clinical trial of point-of-care manufactured fresh anti-CD19/20/22 chimeric antigen receptor T cells for treatment of relapsed or refractory lymphoid malignancies (Non-Hodgkin lymphoma, acute lymphoblastic leukemia, chronic lymphocytic leukemia, B prolymphocytic leukemia). *Blood* 2022;140(Suppl 1):7474–5.
52. Schneider D, Xiong Y, Wu D, Hu P, Alabanza L, Steimle B, et al. Trispecific CD19-CD20-CD22-targeting duoCAR-T cells eliminate antigen-heterogeneous B cell tumors in preclinical models. *Sci Transl Med* 2021;13:eab6401.
53. van de Donk NWCJ, Usmani SZ, Yong K. CAR T-cell therapy for multiple myeloma: state of the art and prospects. *Lancet Haematol* 2021;8:e446–61.
54. Roybal KT, Williams JZ, Morsut L, Rupp LJ, Kolinko I, Choe JH, et al. Engineering T cells with customized therapeutic response programs using synthetic Notch receptors. *Cell* 2016;167:419–32.e16.
55. Kloss CC, Condomines M, Cartellieri M, Bachmann M, Sadelain M. Combinatorial antigen recognition with balanced signaling promotes selective tumor eradication by engineered T cells. *Nat Biotechnol* 2013;31:71–5.
56. Lanitis E, Poussin M, Klattenhoff AW, Song D, Sandaltzopoulos R, June CH, et al. Chimeric antigen receptor T Cells with dissociated signaling domains exhibit focused antitumor activity with reduced potential for toxicity *in vivo*. *Cancer Immunol Res* 2013;1:43–53.
57. Globerson Levin A, Rawet Slobodkin M, Waks T, Horn G, Ninio-Many L, Deshet Unger N, et al. Treatment of multiple myeloma using chimeric antigen receptor T cells with dual specificity. *Cancer Immunol Res* 2020;8: 1485–95.
58. Lajoie MJ, Boyken SE, Salter AI, Bruffey J, Rajan A, Langan RA, et al. Designed protein logic to target cells with precise combinations of surface antigens. *Science* 2020;369:1637–43.
59. Srivastava S, Salter AI, Liggitt D, Yechan-Gunja S, Sarvothama M, Cooper K, et al. Logic-gated ROR1 chimeric antigen receptor expression rescues T cell-mediated toxicity to normal tissues and enables selective tumor targeting. *Cancer Cell* 2019;35:489–503.e8.
60. Richards RM, Zhao F, Freitas KA, Parker KR, Xu P, Fan A, et al. NOT-gated CD93 CAR T cells effectively target AML with minimized endothelial cross-reactivity. *Blood Cancer Discov* 2021;2:648–65.
61. Rischer M, Pscherer S, Duwe S, Vormoor J, Jürgens H, Rossig C. Human gammadelta T cells as mediators of chimeric-receptor redirected anti-tumour immunity. *Br J Haematol* 2004;126:583–92.
62. Harter DC, Simon B, Fujii S-I, Shimizu K, Uslu U, Schuler G, et al. RNA-transfection of γ/δ T cells with a chimeric antigen receptor or an α/β T-cell receptor: a safer alternative to genetically engineered α/β T cells for the immunotherapy of melanoma. *BMC Cancer* 2017;17:551.
63. Wallet MA, Nishimura T, Del Casale C, Lebid A, Salantes B, Santostefano K, et al. Induced pluripotent stem cell-derived gamma delta CAR-T cells for cancer immunotherapy. *Blood* 2021;138(Suppl 1):2771.
64. Neelapu SS, Hamadani M, Miklos DB, Holmes H, Hinkle J, Kennedy-Wilde J, et al. A phase 1 study of ADI-001: anti-CD20 CAR-engineered allogeneic gamma delta ($\gamma\delta$) T cells in adults with B-cell malignancies. *J Clin Oncol* 2022;40(16 suppl):7509.
65. Almeida AR, Correia DV, Fernandes-Platzgummer A, da Silva CL, da Silva MG, Anjos DR, et al. Delta one T cells for immunotherapy of chronic lymphocytic leukemia: clinical-grade expansion/differentiation and preclinical proof of concept. *Clin Cancer Res* 2016;22:5795–804.
66. Jones AB, Rocco A, Lamb LS, Friedman GK, Hjelmeland AB. Regulation of NKG2D stress ligands and its relevance in cancer progression. *Cancers (Basel)* 2022;14:2339.
67. Deng W, Gowen BG, Zhang L, Wang L, Lau S, Iannello A, et al. Antitumor immunity. A shed NKG2D ligand that promotes natural killer cell activation and tumor rejection. *Science* 2015;348:136–9.
68. Ghigo C, Gassard A de, Brune P, Imbert C, Demerle C, Marie-Sarah R, et al. 3 Butyrophilin-3a is expressed in multiple solid tumors: translational research supporting the EVICTION study with ICT01, an anti-BTN3A mAb activating Vg9Vd2 T-cells. *J Immunother Cancer* 2020;8(Suppl 3): A1–A3.
69. Weiss T, Schneider H, Silgner M, Steinle A, Pruschy M, Polić B, et al. NKG2D-Dependent antitumor effects of chemotherapy and radiotherapy against glioblastoma. *Clin Cancer Res* 2018;24:882–95.
70. Baumeister SH, Murad J, Werner L, Daley H, Trebeden-Negre H, Gicobi JK, et al. Phase I trial of autologous CAR T cells targeting NKG2D ligands in patients with AML/MDS and multiple myeloma. *Cancer Immunol Res* 2019; 7:100–12.
71. Weiss T, Weller M, Guckenberger M, Sentman CL, Roth P. NKG2D-Based CAR T cells and radiotherapy exert synergistic efficacy in glioblastoma. *Cancer Res* 2018;78:1031–43.
72. Groh V, Wu J, Yee C, Spies T. Tumour-derived soluble MIC ligands impair expression of NKG2D and T-cell activation. *Nature* 2002;419:734–8.
73. Ferrari de Andrade L, Tay RE, Pan D, Luoma AM, Ito Y, Badrinath S, et al. Antibody-mediated inhibition of MICA and MICB shedding promotes NK cell-driven tumor immunity. *Science* 2018;359:1537–42.
74. Barden M, Holzinger A, Velas L, Mezösi-Csaplár M, Szőör Á, Vereb G, et al. CAR and TCR form individual signaling synapses and do not cross-activate, however, can co-operate in T cell activation. *Front Immunol* 2023;14: 1110482.
75. Hartmann J, Schüssler-Lenz M, Bondanza A, Buchholz CJ. Clinical development of CAR T cells—challenges and opportunities in translating innovative treatment concepts. *EMBO Mol Med* 2017;9:1183–97.
76. Dudley ME, Yang JC, Sherry R, Hughes MS, Royal R, Kammula U, et al. Adoptive cell therapy for patients with metastatic melanoma: evaluation of intensive myeloablative chemoradiation preparative regimens. *J Clin Oncol* 2008;26:5233–9.
77. Pegram HJ, Lee JC, Hayman EG, Imperato GH, Tedder TF, Sadelain M, et al. Tumor-targeted T cells modified to secrete IL-12 eradicate systemic tumors without need for prior conditioning. *Blood* 2012;119:4133–41.
78. Jaspers JE, Khan JF, Godfrey WD, Lopez AV, Ciampricotti M, Rudin CM, et al. IL-18-secreting CAR T cells targeting DLL3 are highly effective in small cell lung cancer models. *J Clin Invest* 2023;133:e166028.
79. Hu B, Ren J, Luo Y, Keith B, Young RM, Scholler J, et al. Augmentation of antitumor immunity by human and mouse CAR T cells secreting IL-18. *Cell Rep* 2017;20:3025–33.
80. Adachi K, Kano Y, Nagai T, Okuyama N, Sakoda Y, Tamada K. IL-7 and CCL19 expression in CAR-T cells improves immune cell infiltration and CAR-T cell survival in the tumor. *Nat Biotechnol* 2018;36:346–51.
81. Duan D, Wang K, Wei C, Feng D, Liu Y, He Q, et al. The BCMA-targeted fourth-generation CAR-T cells secreting IL-7 and CCL19 for therapy of refractory/recurrent multiple myeloma. *Front Immunol* 2021;12:609421.
82. O'Sullivan D, van der Windt GJW, Huang SC-C, Curtis JD, Chang C-H, Buck MD, et al. Memory CD8(+) T cells use cell-intrinsic lipolysis to support the metabolic programming necessary for development. *Immunity* 2014;41: 75–88.
83. Colegio OR, Chu N-Q, Szabo AL, Chu T, Rhebergen AM, Jairam V, et al. Functional polarization of tumour-associated macrophages by tumour-derived lactic acid. *Nature* 2014;513:559–63.
84. Marin E, Bouchet-Delbos L, Renoult O, Louvet C, Nerrière-Daguin V, Managh AJ, et al. Human tolerogenic dendritic cells regulate immune responses through lactate synthesis. *Cell Metab* 2019;30:1075–90.e8.

85. Angelin A, Gil-de-Gómez L, Dahiya S, Jiao J, Guo L, Levine MH, et al. Foxp3 reprograms T cell metabolism to function in low-glucose, high-lactate environments. *Cell Metab* 2017;25:1282–93.e7.
86. Vardhana SA, Hwee MA, Berisa M, Wells DK, Yost KE, King B, et al. Impaired mitochondrial oxidative phosphorylation limits the self-renewal of T cells exposed to persistent antigen. *Nat Immunol* 2020;21:1022–33.
87. Park JH, Kim H-J, Kim CW, Kim HC, Jung Y, Lee H-S, et al. Tumor hypoxia represses $\gamma\delta$ T cell-mediated antitumor immunity against brain tumors. *Nat Immunol* 2021;22:336–46.
88. Sureshbabu SK, Chaukar D, Chiplunkar SV. Hypoxia regulates the differentiation and anti-tumor effector functions of $\gamma\delta$ T cells in oral cancer. *Clin Exp Immunol* 2020;201:40–57.
89. Corsale AM, Di Simone M, Lo Presti E, Picone C, Dieli F, Meraviglia S. Metabolic changes in tumor microenvironment: how could they affect $\gamma\delta$ T cells functions? *Cells* 2021;10:2896.
90. Rodrigues NV, Correia DV, Mensurado S, Nóbrega-Pereira S, deBarros A, Kyle-Cezar F, et al. Low-density lipoprotein uptake inhibits the activation and antitumor functions of human V γ 9V δ 2 T cells. *Cancer Immunol Res* 2018;6:448–57.
91. Zheng W, O'Hear CE, Alli R, Basham JH, Abdelsamed HA, Palmer LE, et al. PI3K orchestration of the in vivo persistence of chimeric antigen receptor-modified T cells. *Leukemia* 2018;32:1157–67.
92. Eikawa S, Nishida M, Mizukami S, Yamazaki C, Nakayama E, Udono H. Immune-mediated antitumor effect by type 2 diabetes drug, metformin. *Proc Natl Acad Sci U S A* 2015;112:1809–14.
93. Hirabayashi K, Du H, Xu Y, Shou P, Zhou X, Fucá G, et al. Dual targeting CAR-T cells with optimal costimulation and metabolic fitness enhance antitumor activity and prevent escape in solid tumors. *Nat Cancer* 2021;2:904–18.
94. Fultang L, Booth S, Yogev O, Martins da Costa B, Tubb V, Panetti S, et al. Metabolic engineering against the arginine microenvironment enhances CAR-T cell proliferation and therapeutic activity. *Blood* 2020;136:1155–60.
95. Ford K, Hanley CJ, Mellone M, Szyndralewicz C, Heitz F, Wiesel P, et al. NOX4 inhibition potentiates immunotherapy by overcoming cancer-associated fibroblast-mediated CD8 T-cell exclusion from tumors. *Cancer Res* 2020;80:1846–60.
96. Kraman M, Bambrough PJ, Arnold JN, Roberts EW, Magiera L, Jones JO, et al. Suppression of antitumor immunity by stromal cells expressing fibroblast activation protein- α . *Science* 2010;330:827–30.
97. Caruana I, Savoldo B, Hoyos V, Weber G, Liu H, Kim ES, et al. Heparanase promotes tumor infiltration and antitumor activity of CAR-redirection T-lymphocytes. *Nat Med* 2015;21:524–9.
98. Ruocco MG, Pilonis KA, Kawashima N, Cammer M, Huang J, Babb JS, et al. Suppressing T cell motility induced by anti-CTLA-4 monotherapy improves antitumor effects. *J Clin Invest* 2012;122:3718–30.
99. Wistuba-Hamprecht K, Martens A, Haehnel K, Geukes Foppen M, Yuan J, Postow MA, et al. Proportions of blood-borne V δ 1+ and V δ 2+ T-cells are associated with overall survival of melanoma patients treated with ipilimumab. *Eur J Cancer* 2016;64:116–26.
100. Casetti R, Agrati C, Wallace M, Sacchi A, Martini F, Martino A, et al. Cutting edge: TGF- β 1 and IL-15 Induce FOXP3+ gammadelta regulatory T cells in the presence of antigen stimulation. *J Immunol* 2009;183:3574–7.
101. Di Stasi A, De Angelis B, Rooney CM, Zhang L, Mahendravada A, Foster AE, et al. T lymphocytes coexpressing CCR4 and a chimeric antigen receptor targeting CD30 have improved homing and antitumor activity in a Hodgkin tumor model. *Blood* 2009;113:6392–402.
102. Stephan SB, Taber AM, Jileeva I, Pegues EP, Sentman CL, Stephan MT. Biopolymer implants enhance the efficacy of adoptive T cell therapy. *Nat Biotechnol* 2015;33:97–101.
103. Smith TT, Moffett HF, Stephan SB, Opel CF, Dumigan AG, Jiang X, et al. Biopolymers codelivering engineered T cells and STING agonists can eliminate heterogeneous tumors. *J Clin Invest* 2017;127:2176–91.
104. Awada H, Paris F, Pecqueur C. Exploiting radiation immunostimulatory effects to improve glioblastoma outcome. *Neuro Oncol* 2023;25:433–46.
105. Fan J, Adams A, Sieg N, Heger J-M, Gödel P, Kutsch N, et al. Potential synergy between radiotherapy and CAR T-cells-a multicentric analysis of the role of radiotherapy in the combination of CAR T cell therapy. *Radiother Oncol* 2023;183:109580.
106. Lamb LS, Bowersock J, Dasgupta A, Gillespie GY, Su Y, Johnson A, et al. Engineered drug resistant $\gamma\delta$ T cells kill glioblastoma cell lines during a chemotherapy challenge: a strategy for combining chemo- and immunotherapy. *PLoS One* 2013;8:e51805.
107. Ma Y, Aymeric L, Locher C, Mattarollo SR, Delahaye NF, Pereira P, et al. Contribution of IL-17-producing gamma delta T cells to the efficacy of anticancer chemotherapy. *J Exp Med* 2011;208:491–503.
108. Soriani A, Zingoni A, Cerboni C, Iannitto ML, Ricciardi MR, Di Galleonardo V, et al. ATM-ATR-dependent up-regulation of DNAM-1 and NKG2D ligands on multiple myeloma cells by therapeutic agents results in enhanced NK-cell susceptibility and is associated with a senescent phenotype. *Blood* 2009;113:3503–11.
109. de Vries NL, van de Haar J, Veninga V, Chalabi M, Ijsselstein ME, van der Ploeg M, et al. $\gamma\delta$ T cells are effectors of immunotherapy in cancers with HLA class I defects. *Nature* 2023;613:743–50.
110. Hegde PS, Karanikas V, Evers S. The where, the when, and the how of immune monitoring for cancer immunotherapies in the era of checkpoint inhibition. *Clin Cancer Res* 2016;22:1865–74.
111. Cherkassky L, Morello A, Villena-Vargas J, Feng Y, Dimitrov DS, Jones DR, et al. Human CAR T cells with cell-intrinsic PD-1 checkpoint blockade resist tumor-mediated inhibition. *J Clin Invest* 2016;126:3130–44.
112. Zumwalde NA, Sharma A, Xu X, Ma S, Schneider CL, Romero-Masters JC, et al. Adoptively transferred V γ 9V δ 2 T cells show potent antitumor effects in a preclinical B cell lymphomagenesis model. *JCI Insight* 2017;2:e93179.